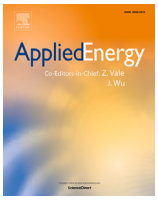
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A smart meter data market framework embedded in electricity market operations

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HIGHLIGHTS

- Propose a data ownership confirmation scheme balancing user privacy and retailer needs.
- Design a data market to monetize data and improve load forecasting.
- Link data volume, forecast accuracy, and electricity-market revenue.
- Develop a benefit allocation method via market clearing and ex-post reallocation.

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ABSTRACT

Modern power systems require extensive data for monitoring, forecasting, economic operation and market clearing. Specifically, smart meter (SM) data can be utilized to analyze consumption patterns and cluster end users, thereby enhancing the accuracy of aggregate load forecasting. However, the ambiguous ownership between various entities and the lack of economic incentives hinder the efficient flow of SM data. To address this issue, this paper proposes an SM data market to promote data trading between end users and the electricity retailer. First, an ownership confirmation scheme is proposed for SM data considering the users' privacy concerns and the retailer's operational requirements, which establishes a foundation for subsequent data market design. Then, an SM data market mechanism accounting for the operation process of the electricity market is designed. End users provide the retailer with access to historical fine-grained SM data and receive payment. The electricity retailer then utilizes fine-grained data to improve aggregate load forecasting and maximize revenue in the electricity market. Finally, a two-stage benefit allocation method for the SM data market is proposed to incentivize participation in data trading. In the first stage, the mapping relationship among the volume of historical data, load forecast accuracy, and expected revenue in the electricity market is quantified to formulate the utility function of stakeholders for data market clearing. In the second stage, the generalized Nash bargaining theory is applied to reallocate the benefits in the SM data market based on realized revenue of the electricity market. Case studies demonstrate that the proposed SM data market achieves optimal benefit allocation under various market conditions and enhances stakeholder participation in data trading.

1. Introduction

Data resources have emerged as a crucial driver of the technological transformation, supporting predictive analytics [1], risk management [2], and decision support [3]. To fully unlock the potential of data resources, promoting data sharing [4] and establishing data markets [5] have become global trends across diverse sectors. For instance, the Data Governance Act enacted by the European Union institutionalizes a harmonized data governance framework to enhance data sharing across sectors and borders [6]. Furthermore, the Chinese government issued the "Opinions on Building a Data-Based System to Better Leverage the Role of Data", which aims to systematically develop a data-based system to effectively eliminate institutional barriers.

In power systems, data is essential for addressing uncertainties in load levels [7], renewable energy generation [1], and electricity prices

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Nomenclature

A. Abbreviations

SM	Smart meter
LSTM	Long short-term memory
MSE	Mean squared error
MIQP	Mixed-integer quadratic programming

B. Sets

$\mathcal{T}$	Set of time periods
$\mathcal{N}$	Set of segments for linearizing the relationship between forecast accuracy and the retailer's revenue increment
$\mathcal{M}$	Set of segments for linearizing the relationship between data volume and forecast accuracy

C. Parameters

p_t	Retail electricity price in period t
π_t	Day-ahead wholesale electricity price in period t
λ_t^+	Positive imbalance price in period t
λ_t^-	Negative imbalance price in period t
α_t^+	Positive imbalance penalty rate in period t
α_t^-	Negative imbalance penalty rate in period t
x_t	Actual electricity load served by the retailer in period t
$f(x_t)$	Probability density function of electricity load in period t
$F(x_t)$	Cumulative distribution function of electricity load in period t
$X_{t,\min}$	Lower bound of the forecast interval with SM data in period t
$X_{t,\max}$	Upper bound of the forecast interval with SM data in period t
$X^0_{t,\min}$	Lower bound of the forecast interval without SM data in period t
$X^0_{t,\max}$	Upper bound of the forecast interval without SM data in period t
$\hat{X}_t$	Load forecast of the retailer in period t
a, b, c	Fitting parameters in the relationship between data volume and forecast accuracy
$Q^{\max}_{data}$	Maximum available volume of SM data
α_{data}	Linear coefficient of end-user privacy-related cost
β_{data}	Quadratic coefficient of end-user privacy-related cost
MSE^j	Breakpoint of MSE associated with segment j
Q^i_{data}	Breakpoint of data volume associated with segment i
Π_t	Retailer revenue in electricity market under uncertain load
Υ^t_{RE}	Mapping function from forecast accuracy interval to expected retailer revenue
n_j	Slope of segment j for linearizing the relationship between forecast accuracy and revenue increment
d_j	Intercept of segment j for linearizing the relationship between forecast accuracy and revenue increment
m_i	Slope of segment i for linearizing the relationship between data volume and forecast accuracy
b_i	Intercept of segment i for linearizing the relationship between data volume and forecast accuracy
M_S	Big-M constant for the segment-selection constraints of MSE.
M_Q	Big-M constant for the segment-selection constraints of Q_{data}.
AR	Additional revenue to be reallocated in the surplus state
AL	Additional loss to be reallocated in the loss state
$\overline{R}_{RE}$	Upper bound of the retailer's realized-revenue fluctuation range
$\underline{R}_{RE}$	Lower bound of the retailer's realized-revenue fluctuation range
α^{Sur}_{EU}	Benefit allocation coefficient of end users in the surplus state
β^{Sur}_{RE}	Benefit allocation coefficient of the retailer in the surplus state
α^{Loss}_{EU}	Loss allocation coefficient of end users in the loss state
β^{Loss}_{RE}	Loss allocation coefficient of the retailer in the loss state
U^0_{EU}	Utility of end users when not participating in ex-post reallocation
U^0_{RE}	Utility of the retailer when not participating in ex-post reallocation

D. Decision variables

Q_{data}	Cleared volume of SM data
ρ_{data}	Clearing price of SM data
$\tilde{X}_t$	Day-ahead electricity procurement quantity in period t
$\tilde{X}^*_t$	Optimal day-ahead electricity procurement quantity in period t
DV_{RE}	Increment of retailer's expected revenue in the electricity market due to the improvement of forecast accuracy
MSE	The mean squared error of load forecasting determined by SM data volume
γ^*_t	Optimal quantile for day-ahead electricity procurement in period t
δ_i	Segment-selection binary variable associated with $\mathcal{M}$
η_j	Segment-selection binary variable associated with $\mathcal{N}$
U_{RE}	Utility of retailer in the market clearing stage
U_{EU}	Utility of end users in the market clearing stage
U^{expost}_{RE}	Utility of retailer in the ex-post reallocation stage
U^{expost}_{EU}	Utility of end users in the ex-post reallocation stage

[8], thereby supporting more reliable operation of power systems. With the growing adoption of the Internet of Things, advanced communication technologies, and smart meters (SMs), the volume of load-side data has increased significantly [9]. The fine-grained SM data from end users offers new insights to improve forecast accuracy and further enhance electricity market revenue [10]. A market-based approach is needed to promote the optimal allocation of SM data, thereby enabling the full utilization of SM data. The development of an SM data market requires clarifying the ownership boundaries of SM data, designing a market operation mechanism for data trading, and establishing a benefit allocation approach among market participants. The following discussion reviews the related studies from these three aspects and identifies the remaining gaps.

Clarifying the ownership boundary of SM data is the prerequisite for developing an SM data market. The EU General Data Protection Regulation [11] and the EU-U.S. Data Privacy Framework [12] mandate the individual's right to data privacy. Concerning load data in power systems, although leveraging fine-grained SM data can bring potential benefits for the retailer or load aggregator [13,14], the analysis of SM data may also expose sensitive socio-demographic information of end users such as occupant details [15], dwelling characteristics [16], and appliance properties [17]. This has raised significant concerns regarding potential privacy breaches among electricity consumers. To address these risks, several practices have been implemented to protect data privacy. Several jurisdictions such as the Netherlands, Norway, the UK and the US have granted consumers the right to opt out of SM installation [18] and set the data collection frequency of digital meters [19] (e.g., monthly). However, the electricity retailer or load aggregator requires access to certain load data for routine operations such as energy billing and demand side management [20]. Additionally, they seek to obtain

additional SM data from users to improve load forecast accuracy and enhance revenue from the electricity market. Consequently, the conflicting interests between end users and the retailer exacerbate the ambiguity regarding the ownership of load data. The unclear delineation of ownership impedes the design of a corresponding market framework.

Market operation mechanisms have been investigated to facilitate the circulation of data resources. A data market model is proposed in [21] to form a sustainable data-sharing coalition and quantify data value for payment. A general data market mechanism is proposed in [22] to support efficient data trading for machine learning tasks. Since data is typically traded for specific business needs, data markets are often coupled with external markets [23]. The above studies fail to integrate data markets with practical scenarios, which limits practical applicability. In the field of power systems, Reference [24] establishes a data market framework, where power system operators purchase historical data from renewable energy suppliers to yield better scheduling strategies. Reference [25] proposes a data market to incentivize collaboration between different renewable energy agents in a given region. Reference [26] investigates the effect of data on the operators' utility function in the energy balancing market. The derived equilibrium maximizes social welfare and minimizes total mismatch. However, extending these ideas to SM data entails additional considerations. First, SM data is supplied by a large population of end users, where data is abundant and ownership is complex. Second, the economic value of SM data is realized through electricity market revenue rather than through forecast accuracy alone. Third, since the economic value of SM data is ultimately realized through electricity market revenue, existing data market studies have not sufficiently considered how the operation of the data market should be coordinated with the operational process of the electricity market.

Benefit allocation approaches have been widely studied to share the economic value created by data among participants. The existing research can be divided into two categories from the perspective of application scenarios: raw data trading and forecast information trading. The first focuses on using raw data from end users to improve model performance. The second focuses on allocating revenue of forecast information. In raw data trading, a regression market framework was proposed in [27] to support agents in sharing their data resources with a central agent to train an energy forecast model with higher accuracy. The attributes of uncertainty, timeliness, and completeness are utilized to realize multidimensional electricity data valuation in [28]. Reference [29] proposes a data market framework that incorporates data collection costs and forecast accuracy rewards, facilitating the sharing of raw data among electricity market participants. In the forecast information trading, Reference [30] investigates the impact of various forecast information on the operational cost of power systems. A novel data valuation approach is proposed in [31], which depicts the quantitative relationships between the reduced forecast uncertainty and increased revenue for the electricity retailer. In [32], a profit allocation approach for forecast information sharing is proposed to distribute profits between the electricity retailer and data suppliers.

Although the aforementioned studies are highly innovative, several limitations remain. First, the studies in [27–29] quantify the effect of raw data on forecast accuracy, thereby providing a basis for data valuation. However, these studies do not fully characterize the subsequent value realization process from forecast accuracy improvement to electricity market revenue. Second, the studies in [30–32] represent a specific form of forecast information market research, focusing mainly on the effect of forecast accuracy on electricity market revenue. However, these studies usually take forecast information as the tradable object and do not sufficiently consider the underlying relationship between data volume and forecast accuracy, or the cost associated with high-precision forecast information. Therefore, existing studies on data valuation and forecast information markets have not yet fully established the value-cost chain from data volume to forecast accuracy and further to electricity market revenue. A related study in [33] employs cooperative game theory to monetize customer load data by linking raw data, forecast accuracy, and economic revenue. However, it does not establish a dedicated data market mechanism integrated with the operational timeline of existing electricity markets, thereby limiting its practical applicability.

Therefore, this paper proposes a load-side SM data market framework that promotes the data trading between end users and the electricity retailer. The main contributions are threefold.

1. A confirmation scheme for load data ownership is proposed. By analyzing the relevant legislation and implementation practices, the data ownership is assigned across temporal-spatial dimensions to balance the privacy needs of end users and the retailer's operational requirements. This scheme serves as a pre-market rule for the subsequent SM data market.
2. A market mechanism is developed for SM data. Unlike existing studies, a load-side scenario is investigated where data ownership is ambiguous. On this basis, the proposed mechanism is integrated with the operation process of wholesale market and retail market in power systems to facilitate the circulation of SM data.
3. A two-stage benefit allocation approach is designed for the SM data market. Unlike previous research, the relationship among SM data volume, forecast accuracy, and retailer's expected revenue in the electricity market is analyzed to quantify the value-cost characteristics of SM data for formulating the market clearing problem. Furthermore, an ex-post reallocation approach based on generalized Nash bargaining theory is incorporated to adjust benefit allocation in the SM data market.

2. The ownership confirmation scheme between end users and the retailer

2.1. Privacy concerns of end users' smart meter data

The emergence of SMs has changed the perception of user privacy in electricity consumption data. Before SMs were widely adopted, electricity usage information was mainly collected for monthly billing, and privacy concerns related to such data received relatively limited attention. With the large-scale deployment of SMs, electricity consumption can be recorded at 15-minute or 30-minute intervals, making it possible to infer more detailed user behaviors and increasing the risk of privacy breaches. For instance, analyzing fine-grained historical load data from SMs can reveal sensitive socio-economic information, such as employment status, household composition, housing type, and retirement status.

In recent years, there has been increasing attention to the ownership of SM data. The Fairness in Information Practices Principles and the EU General Data Protection Regulation affirm that individuals have the right to refuse data collection, underscoring this as a fundamental component of user privacy rights. Similarly, the 2018 California Consumer Privacy Act grants individuals the right to know what types of information are collected, the right to opt out of data sales to third parties, and the right to delete their personal information.

2.2. The retailer's motivation for accessing load data

The grid operator and other relevant entities participate in the construction of grid infrastructure and are responsible for ensuring the secure operation of the grid. As a result, they have access to substantial data resources from the load side. In routine operation, load data is critical for grid management. For instance, historical load data of grid nodes are used to forecast load profiles, while data collected from electricity meters serve as the foundation for billing and demand-side management.

Fine-grained SM data offers valuable information for improving load forecast accuracy, thereby enhancing retailers' revenue in the electricity market. The clustering algorithm is adopted to analyze the fine-grained SM data to group different end users based on consumption patterns.

Then, tailored models are applied to forecast the load profile of each group. The aggregate load forecast is obtained by summing the forecasts of all groups.

2.3. Current practices in data confirmation and the proposed data ownership scheme

Many countries have implemented measures to safeguard user privacy related to SM data. The right to opt out of SM installation is already practiced in the Canadian provinces of British Columbia and Quebec, the Netherlands, the UK, and certain U.S. states [18,34,35]. In these regions, end users have the option to retain an old analog meter or impose restrictions on SM data collection, such as mandating manual readings or limiting the frequency of data collection. However, end users' decisions to opt out may adversely affect the retailer's interests. If a significant number of end users choose to opt out, the retailer may lose revenue from refined energy management such as load management, time-of-use pricing, and transactive energy.

Based on the above analysis, the proposed scheme differentiates the SM data along temporal and spatial dimensions. This approach is different from conventional hierarchical data-collection arrangements. In the conventional arrangements, end-user data is typically assumed to be available to utilities or aggregators once collected through metering infrastructure. Instead, this approach focuses on a setting where access to fine-grained individual SM data cannot be taken as naturally granted. As illustrated in Fig. 1, fine-grained data at the individual level refers to second-level, minute-level, or hourly-level power sequences measured by a smart meter. The analysis of such data can reveal detailed consumption patterns and may raise privacy concerns. Coarse-grained data at the individual level refers to daily, weekly, or monthly electricity consumption records, which are mainly required for electricity billing. Fine-grained data at the aggregated level refers to node-level power sequences at short time intervals, which support the analysis of power system operations and retailers' electricity procurement. Coarse-grained data at the aggregated level refers to node-level daily or monthly electricity consumption, which is used for demand trend analysis and power network planning. Fine-grained data at the individual level is assigned to end users. The remaining data is assigned to the retailer. Consequently, the fine-grained SM data at the individual level (indicated by the lower-left area in Fig. 1) can be traded between end users and the retailer through the proposed data market framework.

It should be clarified that data trading discussed in this paper does not involve the transfer of data ownership. Instead, it refers to the authorization of the rights to use data. In other words, end users retain the ownership of their fine-grained SM data, while the retailer obtains the right to use the authorized data for specified operational purposes, such as load forecasting and decision-making in electricity markets. Accordingly, subsequent references to data trading should be understood in this sense.

Data trading under this scheme is voluntary. End users who authorize the use of fine-grained data can participate in the SM data market and receive payment, while users who do not provide such authorization remain outside the trading mechanism. Therefore, incomplete participation does not invalidate the framework itself. The practical implementation of this scheme relies on both governance rules and technical support. In particular, the use of fine-grained SM data should be based on explicit user authorization, and purpose limitation, which can be supported by access control, encryption, and traceable logging. This ownership confirmation is introduced as a precondition for the subsequent market mechanism design and two-stage benefit allocation.

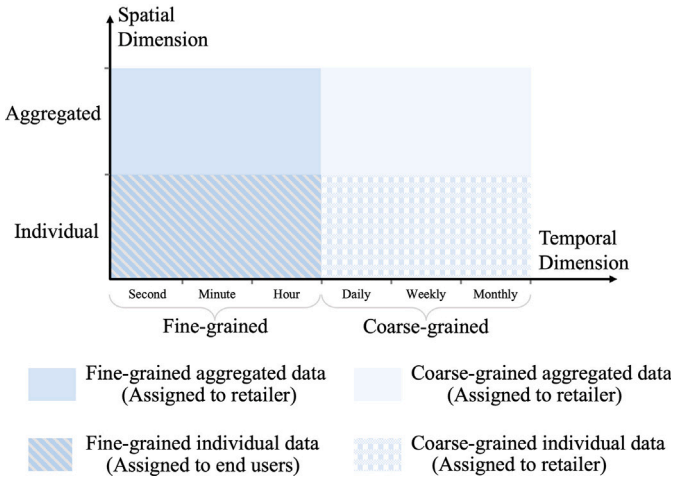


Fig. 1. The proposed ownership confirmation for load data.

3. Smart meter data market mechanism

3.1. Smart meter data for load forecasting

A cluster-based load forecasting method is used to exploit fine-grained SM data for improving aggregate load forecast accuracy. Historical fine-grained SM data contributed by participating end users is first grouped using a clustering method such as k-means to identify users with similar consumption patterns. For each cluster, a forecasting model is trained using historical load sequences, and the aggregate load forecast is obtained by summing the forecasts across all clusters. In the case study, an LSTM-based forecasting model is adopted. The corresponding model settings, data split, and training procedure of the cluster-based load forecasting method are summarized in Appendix A.

By analyzing fine-grained SM data, the electricity retailer can gain insights into the consumption patterns of an individual user. This improves aggregate load forecasting accuracy. In the proposed framework, fine-grained SM data is used as the forecasting input. By utilizing the fine-grained SM data, the retailer groups users with similar consumption patterns and obtains a more accurate load forecast. A more accurate forecast reduces the uncertainty faced by the retailer in electricity-market decisions. The market-clearing model determines the traded data quantity and the corresponding data price. The other three data categories in Fig. 1 are not included in the proposed SM data market.

3.2. The stakeholders in smart meter data market

Retailer: The electricity retailer engages in both wholesale and retail markets to generate revenue. The wholesale market follows a two-settlement structure, comprising a day-ahead market and an intraday balancing market. This market mechanism is commonly used in many countries and regions (e.g., the U.S. and Iberia). In the day-ahead market, the retailer purchases electricity according to load forecast information. In the intraday balancing market, the retailer needs to pay imbalance costs due to electricity deviations. In this context, load forecast inaccuracy is unavoidable, potentially resulting in imbalance costs and reduced system reliability. To mitigate these risks and maximize profitability, the electricity retailer seeks accurate load forecast information. By obtaining fine-grained SM data from end users and utilizing the aforementioned clustering-based load forecast method, the retailer generates high-accuracy load forecast information. In this framework, the retailer is modeled as an economically motivated market participant rather than a non-profit coordinator. The retailer uses fine-grained SM data from end users to improve aggregate load forecasting and support decision-making in the electricity market. Therefore, the retailer acts as the entity that links data trading to electricity market.

End users: End users act as providers of SM data in the SM data market. They provide the retailer with fine-grained SM data for load forecasting and receive payment in return. Meanwhile, the retailer is required to ensure that the SM data is not used beyond the agreed scope.

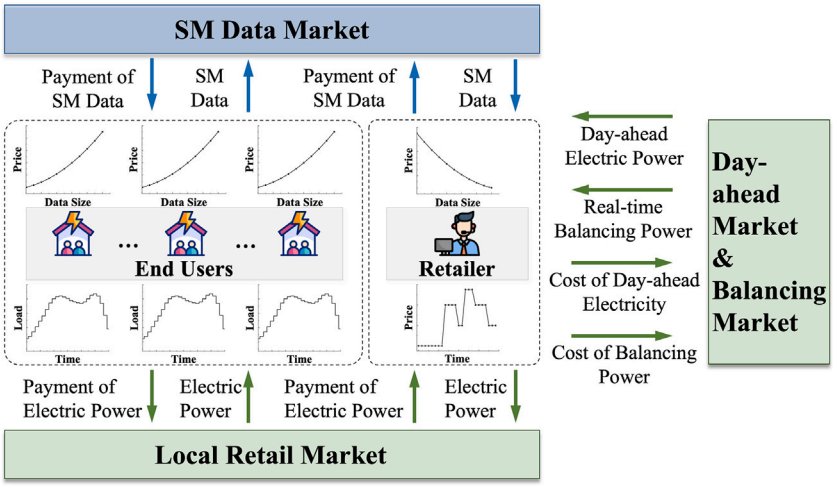


Fig. 2. The framework of SM data market.

In summary, end users participate in the SM data market by providing the retailer with access to historical SM data for electricity-market decision-making, thereby obtaining additional economic benefits.

The proposed SM data market framework is illustrated in Fig. 2.

3.3. The operation process of smart meter data market embedded in the electricity market

Fig. 3 illustrates the operational process of the proposed SM data market, with the detailed steps described below.

Step 1 to Step 4 pertain to the day-ahead operation of the SM data market. End users and the retailer submit willingness to sell or buy the right to use SM data, respectively. Based on the clearing results, end users grant the retailer access to fine-grained SM data, while retailer provides payment to end users accordingly. Notably, before submitting bid curves, the retailer needs to estimate the relationship between data volume and forecast accuracy using a small sample of fine-grained SM data provided by end users. Then, by estimating the day-ahead wholesale electricity price and the intraday imbalance price, the economic value of SM data can be quantified. In this stage, the retailer validates candidate SM data by comparing forecast accuracy with and without it. Only SM data that yields a measurable reduction in forecast error is retained for trading. SM data that does not improve accuracy is excluded.

Step 5 to Step 7 pertain to the trading process in the day-ahead wholesale electricity market. Based on the fine-grained SM data obtained in the data market, the retailer generates high-accuracy load forecast information and subsequently makes optimal decisions in the day-ahead market to procure the required electricity.

Step 8 to Step 9 involve the trading process in the intraday balancing market. Due to the inherent uncertainty of load, the deviation between day-ahead load forecasts and actual load level is unavoidable. Consequently, the retailer must procure balancing power from the intraday balancing market to offset load fluctuations.

Step 10 pertains to the ex-post reallocation in the SM data market. The inherent uncertainty of load level and electricity market price introduces inaccuracies in the estimation of the economic value of SM data. Therefore, it is necessary to redistribute benefits among participants based on the final outcome of electricity market revenue.

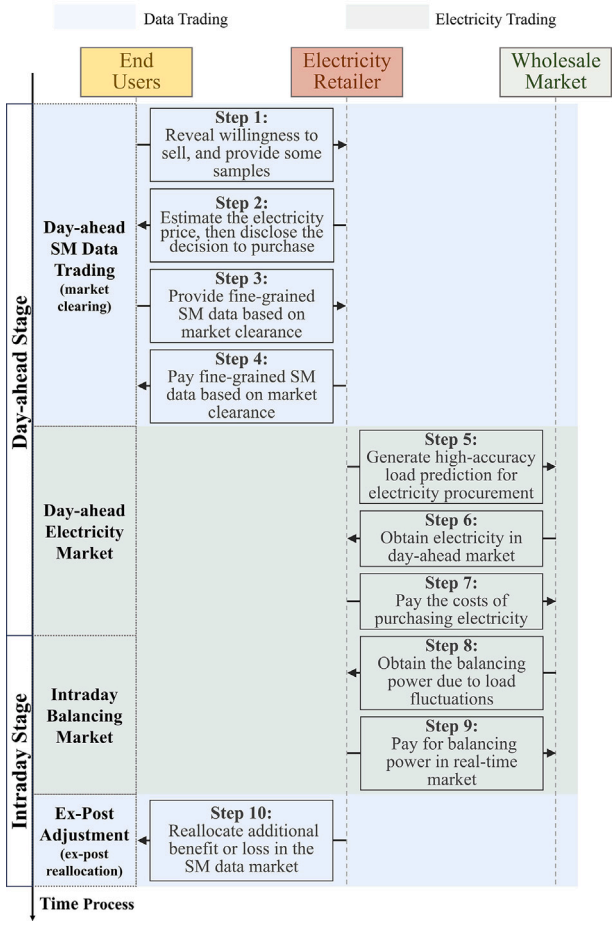


Fig. 3. The operational process of SM data market.

4. Mathematical model for benefit allocation

After establishing the data market framework outlined above, benefit allocation among participants is required to incentivize participation in data sharing. The proposed benefit allocation framework consists of two stages: market clearing and ex-post reallocation. In the stage of market clearing, the trading outcome is determined based on the value-cost characteristics of SM data. In the stage of ex-post reallocation, the realized surplus or loss is redistributed after the electricity market outcome is observed.

In the proposed model, only fine-grained SM data is included. The fine-grained SM data improves load forecasting accuracy and thereby affects the retailer's expected revenue in electricity market. The coarse-grained data at the individual level, the fine-grained data at the

aggregated level, and the coarse-grained data at the aggregated level shown in Fig. 1 are not traded in the proposed framework.

4.1. Modeling the objective function for end users

End users act as suppliers in the SM data market by allowing fine-grained SM data to be used for forecasting and market decision-making. The retailer provides payment for such data use. For end users, participation in the SM data market introduces privacy-related cost. The privacy-related cost is modeled as an increasing and convex function of the fine-grained SM data volume. This assumption is consistent with the literature basis. Reference [36] shows that the cost associated with data can increase convexly with the volume of data, indicating that a larger volume of data may involve progressively higher costs. Reference [37] provides a related model by representing privacy loss as a quadratic function of the proportion of personal data used. In this paper, a quadratic function is adopted to represent this characteristic and maintain the tractability of the market-clearing problem. The utility function of end users is expressed as follows.

$$U_{EU} = \rho_{data} Q_{data} - \alpha_{data} Q_{data} - \beta_{data} \left(Q_{data}\right)^2 \tag{1}$$

In practice, α_{data} and β_{data} can be calibrated from user preferences, such as survey-elicited required payments under different levels of fine-grained SM data use, or inferred from observed participation decisions in pilot programs.

Fine-grained SM data reveals private information about end users, such as household routines, occupancy patterns, and appliance use. However, such private information is not included in the proposed market. Only fine-grained SM data is used for load forecasting and market clearing. With the utilization of fine-grained SM data, the retailer applies a clustering-based load forecasting approach to forecast aggregate load. The resulting improvement in forecast accuracy can further increase the retailer's revenue in the electricity market. The detailed process of the clustering-based aggregate forecasting method used in this paper is provided in Appendix A.

4.2. Modeling the objective function for retailer

The electricity retailer purchases electricity from the wholesale market and sells electricity to end users in the retail market. The revenue in electricity market is derived from the difference between retail revenue and wholesale costs. Here, stochastic programming is used to model decision-making in electricity market, and privacy protection is addressed through ownership confirmation and controlled authorization of fine-grained SM data. Thus, the electricity market revenue of retailer can be analyzed as follows.

$$\Pi_t = \begin{cases} p_t x_t - \pi_t \tilde{X}_t + \lambda_t^+ \left(\tilde{X}_t - x_t\right), & X_{t,\min} \le x_t \le \tilde{X}_t \\ p_t x_t - \pi_t \tilde{X}_t - \lambda_t^- \left(x_t - \tilde{X}_t\right), & \tilde{X}_t \le x_t \le X_{t,\max} \end{cases} \tag{2}$$

$$\begin{cases} \lambda_t^- = \left(1 + \alpha_t^-\right) \pi_t \\ \lambda_t^+ = \left(1 - \alpha_t^+\right) \pi_t \end{cases} \tag{3}$$

The decision-making strategy of the retailer in the electricity market can be expressed as a stochastic optimization problem.

$$\begin{aligned} \max \mathbb{E}\left(\sum_{\forall t \in \mathcal{T}} \Pi_t\left(\pi_t, \lambda_t^-, \lambda_t^+\right)\right) &= \sum_{\forall t \in \mathcal{T}} \int_{X_{t,\min}}^{X_{t,\max}} \Pi_t\left(x_t\right) f_{X_t}\left(x_t\right) \mathrm{d}x_t \\ &= \sum_{\forall t \in \mathcal{T}} \left(\int_{X_{t,\min}}^{\tilde{X}_t} \left[p_t x_t - \pi_t \tilde{X}_t + \lambda_t^+ \left(\tilde{X}_t - x_t\right)\right] f(x_t) \mathrm{d}x_t \right. \\ &\quad \left. + \int_{\tilde{X}_t}^{X_{t,\max}} \left[p_t x_t - \pi_t \tilde{X}_t - \lambda_t^- \left(x_t - \tilde{X}_t\right)\right] f(x_t) \mathrm{d}x_t \right) \end{aligned} \tag{4}$$

$$\text{s.t.: } X_{t,\min} \le \tilde{X}_t \le X_{t,\max}$$

where $f(x_t)$ is the probability density function of electricity load. In the proposed model, the actual load is modeled as an uncertain parameter around the forecast value, and the forecast error is assumed to follow a Gaussian distribution. The imbalance settlement is represented by an exogenous asymmetric pricing rule. Specifically, the shortage imbalance price is set higher than the wholesale price, while the excessive imbalance price is set lower than the wholesale price. Under this setting, the retailer's decision-making model follows a newsvendor-type problem between under-procurement and over-procurement. According to the first-order optimality condition, the optimal quantile for electricity procurement and the corresponding optimal procurement amount in the day-ahead market are as follows. The detailed derivation process can be found in Appendix B.

$$\gamma_t^* = \frac{\lambda_t^- - \pi_t}{\lambda_t^- - \lambda_t^+}, \quad \forall t \in \mathcal{T} \tag{5}$$

$$\tilde{X}_t^* = F^{-1}\left(\frac{\lambda_t^- - \pi_t}{\lambda_t^- - \lambda_t^+}\right) = F^{-1}\left(\gamma_t^*\right), \quad \forall t \in \mathcal{T} \tag{6}$$

This optimal quantile-based bidding strategy follows the framework originally proposed in [38] for wind power producers. A deterministic calculation can be interpreted as the retailer's market outcome under a given load forecasting result. If such deterministic calculations are repeated over multiple realizations and then averaged, the value can approximate the expected outcome obtained from stochastic programming. However, the deterministic averaging is only an ex-post numerical approximation and does not directly characterize the economic value of SM data under uncertainty. Since the proposed framework aims to evaluate how the SM data volume affects the retailer's expected electricity-market revenue through forecast-uncertainty reduction, the stochastic formulation remains the main tool for economic evaluation of SM data.

By substituting Eq. (6) into Eq. (4), the retailer's expected revenue can be analytically expressed over the interval of load forecast $\left[X_{t,\min}, X_{t,\max}\right]$.

$$\begin{aligned} \Upsilon_{RE}^t(X_{t,\min}, X_{t,\max}) &= \left(p_t - \lambda_t^+\right) \int_{X_{t,\min}}^{\tilde{X}_t^*} x_t f(x_t) \mathrm{d}x_t \\ &\quad + \left(\lambda_t^+ \tilde{X}_t^* - \pi_t \tilde{X}_t^*\right) F(x_t)\Big|_{X_{t,\min}}^{\tilde{X}_t^*} \\ &\quad + \left(p_t - \lambda_t^-\right) \int_{\tilde{X}_t^*}^{X_{t,\max}} x_t f(x_t) \mathrm{d}x_t \\ &\quad + \left(\lambda_t^- \tilde{X}_t^* - \pi_t \tilde{X}_t^*\right) F(x_t)\Big|_{\tilde{X}_t^*}^{X_{t,\max}} \end{aligned} \tag{7}$$

For each $\left[X_{t,\min}, X_{t,\max}\right]$, calculate the corresponding value of expected revenue, and then fit the results to obtain the analytical mapping relationship between forecast accuracy and retailer's expected revenue.

$$\mathbb{E}\left(\sum_{\forall t \in \mathcal{T}} \Pi_t\right) = \sum_{\forall t \in \mathcal{T}} \Upsilon_{RE}^t(X_{t,\min}, X_{t,\max}) \tag{8}$$

Based on the retailer's original forecast interval $[X_{t,\min}^0, X_{t,\max}^0]$, the improvement in retailer's revenue in the electricity market is obtained by subtracting the revenue corresponding to the original forecast accuracy from the revenue under different forecast accuracy.

$$DV_{RE} = \sum_{\forall t \in \mathcal{T}} \left[\Upsilon_{RE}^t(X_{t,\min}, X_{t,\max}) - \Upsilon_{RE}^t(X_{t,\min}^0, X_{t,\max}^0)\right] \tag{9}$$

where DV_{RE} reflects the increase in the electricity market revenue, i.e., the economic value associated with the corresponding SM data or forecast accuracy.

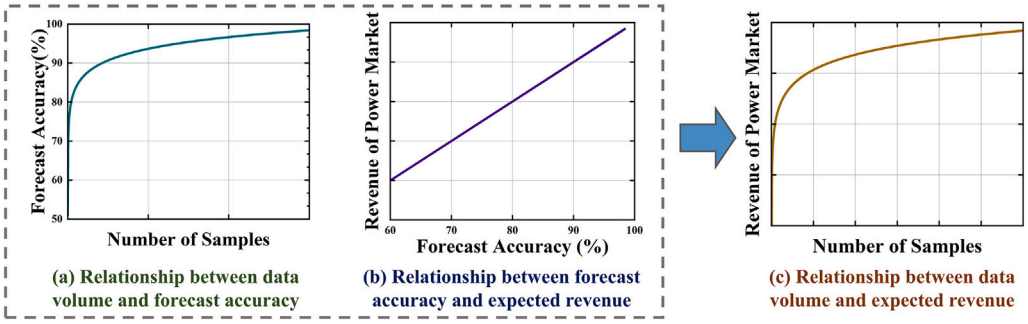


Fig. 4. Derivation of the relationship between data volume and retailer's expected revenue in the electricity market.

The relationship between SM data volume and forecast accuracy is characterized by the following power-law function.

$$MSE = \left(\frac{a}{Q_{data} - c}\right)^{\frac{1}{b}} \tag{10}$$

where a, b, and c are fitting parameters, and MSE denotes the mean squared error. In this study, these parameters are obtained by fitting the numerical results in the case study. Eq. (10) is an empirical approximation used to characterize the marginal improvement in forecast accuracy as data volume increases. This fitted relationship is also consistent with related studies [38].

Under the assumption of Gaussian zero-mean errors, the relationship between the forecast accuracy in Eq. (9) and MSE in Eq. (10) is as follows, with the detailed derivation provided in Appendix C.

$$\begin{cases} X_{t,\min} = \hat{X}_t - 3\sqrt{MSE}, \\ X_{t,\max} = \hat{X}_t + 3\sqrt{MSE}. \end{cases} \tag{11}$$

Fig. 4 illustrates how data volume affects forecast accuracy and consequently impacts revenue in electricity market. In this way, this figure provides an intuitive explanation of the interaction between the SM data market and the electricity market. Although the curves shown in Fig. 4 are schematic, the qualitative trends are consistent with the empirical relationships obtained in the subsequent numerical analysis. This figure is intended to clarify the derivation logic linking the SM data market and the electricity market.

By combining Eqs. (8) and (10), the relationship between the retailer's revenue in the electricity market and the fine-grained SM data volume can be derived. Subtracting the payment for obtaining the SM data, the retailer's objective function in the SM data market is obtained as follows.

$$U_{RE} = DV_{RE} - \rho_{data} Q_{data} \tag{12}$$

4.3. Market clearing problem

The market clearing is performed by maximizing the social welfare in the SM data market, aiming to achieve optimal allocation of SM data. The problem of data market clearing is as follows.

$$\max(U_{RE} + U_{EU}) = DV_{RE} - \alpha_{data} Q_{data} - \beta_{data} (Q_{data})^2 \tag{13}$$

s.t.: Eqs. (8)–(10)

$$0 \le Q_{data} \le Q_{data}^{\max} \tag{14}$$

$$X_{t,\min} \ge X_{t,\min}^0, X_{t,\max}^0 \ge X_{t,\max}, \quad \forall t \in \mathcal{T} \tag{15}$$

$$\rho_{data} = \frac{d(DV_{RE})}{d(Q_{data})} \tag{16}$$

The market clearing is formulated under a non-strategic setting. The clearing outcome is determined by the value-cost characteristics of SM data, as reflected by the bid and offer curves. In practical implementation, strategic behavior and market power may affect the clearing outcome. For example, end users may adjust their offer prices to seek higher payment, while a retailer may also adjust bid curve to influence the clearing price. Such behaviors may shift the offer or bid curves and change the clearing quantity and price. Strategic behavior and market power are not modeled in this paper, because the current study focuses on the value-cost characterization of SM data and the corresponding welfare-based market clearing. Extending the framework to strategic bidding and market-power analysis is an important direction for future work.

Eq. (9) introduces a non-convex relationship between forecast accuracy and the retailer's expected revenue increment, and Eq. (10) introduces a non-convex relationship between data volume and forecast error. The piecewise linearized approach is adopted to obtain a tractable reformulation.

For Eq. (9), the piecewise linearized constraints are presented as follows.

$$n_j = \frac{DV_{RE}\left(MSE^{j+1}\right) - DV_{RE}\left(MSE^{j}\right)}{MSE^{j+1} - MSE^{j}}, \forall j \in \mathcal{N} \tag{17}$$

$$d_j = DV_{RE}\left(MSE^{j}\right) - n_j MSE^{j}, \forall j \in \mathcal{N} \tag{18}$$

$$\sum_{j \in \mathcal{N}} \eta_j = 1 \tag{19}$$

$$MSE^{j} - M_S(1 - \eta_j) \le MSE \le MSE^{j+1} + M_S(1 - \eta_j), \quad \forall j \in \mathcal{N} \tag{20}$$

For Eq. (10), the piecewise linearization constraints are presented as follows.

$$m_i = \frac{MSE^{i+1} - MSE^{i}}{Q_{data}^{i+1} - Q_{data}^{i}}, \quad \forall i \in \mathcal{M} \tag{21}$$

$$b_i = MSE^{i} - m_i Q_{data}^{i}, \quad \forall i \in \mathcal{M} \tag{22}$$

$$\sum_{i \in \mathcal{M}} \delta_i = 1 \tag{23}$$

$$Q_{\text{data}}^{i} - M_Q(1 - \delta_i) \le Q_{\text{data}} \le Q_{\text{data}}^{i+1} + M_Q(1 - \delta_i), \quad \forall i \in \mathcal{M} \tag{24}$$

Based on the above transformation, Eq. (16) can be reformulated as follows.

$$\rho_{data} = \sum_{j \in \mathcal{N}} \eta_j n_j \times \sum_{i \in \mathcal{M}} \delta_i m_i \tag{25}$$

Thus, the original optimization problem is transformed into a mixed-integer quadratic programming (MIQP) problem. As a standard optimization form supported by modern commercial solvers, the resulting MIQP can be directly solved using Gurobi. Under the adopted solver settings, the model reaches the optimal solution within the prescribed relative optimality-gap tolerance. This indicates that the reformulated model can be solved reliably.

4.4. Ex-post reallocation problem

The economic value of SM data is characterized based on the retailer's expected revenue. However, the realized revenue may deviate from the expected value due to load uncertainty and electricity price fluctuations. Therefore, the ex-post reallocation mechanism is introduced to address large revenue deviations caused by substantial forecast errors or unexpected price shocks. The procedure is as follows: (1) simulate multiple load scenarios to obtain the retailer's realized revenue in the electricity market; (2) construct a probabilistic range $[\underline{R}_{RE}, \overline{R}_{RE}]$ using the 5th and 95th percentiles; and (3) conduct no reallocation if the realized revenue falls within this range, and conduct ex-post reallocation if the realized revenue falls outside this range.

4.4.1. Surplus state

This state occurs when the retailer's revenue in the electricity market exceeds the upper bound of the fluctuation range. Such a surplus may arise from a small forecast error or a lower actual electricity price than the estimated value. In this state, the retailer gains additional revenue, whereas the additional revenue is not allocated to end users under the market clearing outcome. Therefore, an ex-post reallocation for the additional revenue between the retailer and end users is required. The generalized Nash bargaining approach is employed for modeling the optimization problem in this state.

$$\max \left(U_{EU}^{expost} - U_{EU}^{0}\right)^{\alpha_{EU}^{Sur}} \left(U_{RE}^{expost} - U_{RE}^{0}\right)^{\beta_{RE}^{Sur}} \tag{26}$$

s.t.:

$$U_{EU}^{expost} + U_{RE}^{expost} - U_{EU}^{0} - U_{RE}^{0} - \bar{R}_{RE} = AR \tag{27}$$

$$U_{EU}^{expost} \geq U_{EU}^{0} \tag{28}$$

$$U_{RE}^{expost} \geq U_{RE}^{0} \tag{29}$$

where α_{EU}^{Sur} and β_{RE}^{Sur} represent the benefit allocation coefficient of end users and the retailer in this state, respectively. These coefficients represent the relative bargaining power of the two parties in allocating the additional realized benefit. A larger coefficient indicates greater bargaining power and therefore leads to a larger share of the surplus. In the following case study, the coefficient settings are selected to provide a reasonable representation of possible market conditions. For example, one coefficient setting places greater emphasis on incentivizing user participation in an early-stage data market, while another may place relatively greater emphasis on operation sustainability. U_{EU}^{0} and U_{RE}^{0} represent the utility values of end users and the retailer when they do not participate in the ex-post reallocation under the surplus state, respectively. This method assumes that if end users and the retailer opt out of the ex-post reallocation, they will be unable to share the additional revenue. Consequently, U_{EU}^{0} and U_{RE}^{0} are equal to zero, respectively.

4.4.2. Loss state

This state occurs when the retailer's revenue in the electricity market falls below the lower bound of the fluctuation range. Such a loss may arise from a large forecast error or a higher actual electricity price than the estimated value. In this state, the retailer incurs economic losses, leading to an imbalance in benefit distribution within the SM data market, which is detrimental to its long-term operation. Therefore, an ex-post reallocation of economic losses between the retailer and end users is necessary. The generalized Nash bargaining approach is employed for modeling the optimization problem in this state.

$$\min \left(U_{EU}^{0} - U_{EU}^{expost}\right)^{\alpha_{EU}^{Loss}} \left(U_{RE}^{0} - U_{RE}^{expost}\right)^{\beta_{RE}^{Loss}} \tag{30}$$

s.t.:

Table 1
Statistical characteristics of the processed dataset.

Statistical indicator	Value
Mean hourly load (kW)	3284.07
Hourly load standard deviation (kW)	1992.57
Hourly load variation coefficient	0.60
Hourly 5th percentile (kW)	863.69
Hourly 95th percentile (kW)	7141.38
Mean daily consumption (kWh/day)	78,817.69
Daily consumption standard deviation (kWh/day)	30,398.87
Hourly peak-to-valley ratio	19.42
Load factor	0.27
Mean absolute hourly ramping (kW)	584.59

$$U_{EU}^{expost} + U_{RE}^{expost} - U_{EU}^{0} - U_{RE}^{0} - \underline{R}_{RE} = AL \tag{31}$$

$$U_{EU}^{0} \geq U_{EU}^{expost} \tag{32}$$

$$U_{RE}^{0} \geq U_{RE}^{expost} \tag{33}$$

where α_{EU}^{Loss} and β_{RE}^{Loss} represent the loss allocation coefficient of end users and the retailer in this state, respectively. These coefficients reflect the relative bargaining power of the two parties to bear additional cost in the ex-post stage. In general, a smaller coefficient corresponds to a lower share of the allocated loss. U_{EU}^{0} and U_{RE}^{0} represent the utility values of end users and the retailer when they do not participate in the ex-post reallocation process under the loss state, respectively. This method assumes that if end users and the retailer opt out of the ex-post loss redistribution, neither party bears the additional loss. Consequently, U_{EU}^{0} and U_{RE}^{0} are equal to zero, respectively.

5. Case study

5.1. Dataset overview

The proposed framework is validated on a real-world SM dataset, namely *CER Smart Metering Project - Electricity Customer Behavior Trial (CBT)* dataset. Released by the Irish energy regulator, this dataset contains 30-minute SM readings from a large number of residential and small enterprise customers, covering Sep. 2009-Jan. 2011. Standard preprocessing steps are performed, including (i) removal of missing/duplicated timestamps, (ii) outlier screening, and (iii) temporal alignment to a uniform time resolution. The main statistical characteristics of the processed dataset for validation are summarized in Table 1.

5.2. Scenario parameter setting and justification

The case study is constructed under a single-retailer setting, where one retailer serves a group of end users. Accordingly, the relevant information exchange takes place only between end users and the retailer. During market clearing, end users and the retailer submit data-trading information related to data volume and price to the SM data market. After market clearing, participating end users provide the corresponding fine-grained data to the retailer, and the retailer provides payment in return. The forecasting experiments are conducted at an hourly resolution.

Following [29], the retail price p_t is set to 72.9 €/MWh and the wholesale price π_t is set to 66.5 €/MWh. To investigate the proposed framework under diverse operating conditions, a set of scenarios is constructed by varying (i) the imbalance penalty rate (α_t^+, α_t^-) and (ii) the SM data-cost coefficient β_{data}. For clarity, the considered scenarios are divided into two groups according to the varying parameters: scenarios with different electricity-market settings (Scenarios 1–4) and scenarios with different data-cost settings (Scenarios 5–8).

Scenarios with different electricity-market settings

Scenario 1 (S1): $\alpha_t^+ = 0.3$, $\alpha_t^- = 0.3$

Scenario 2 (S2): $\alpha_t^+ = 0.4$, $\alpha_t^- = 0.4$

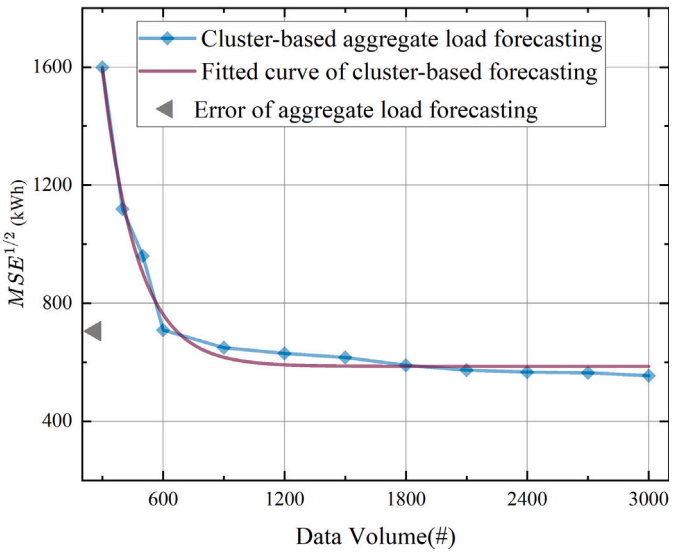


Fig. 5. The relationship between data volume and load forecasting error under the cluster-based load forecasting method.

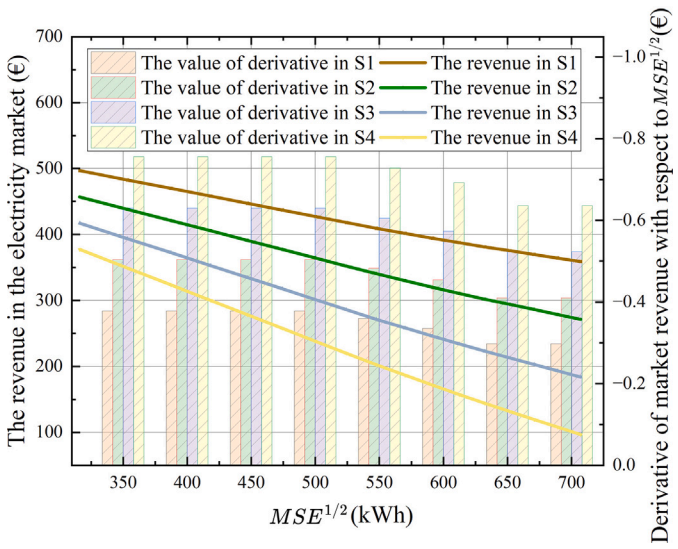


Fig. 6. The impact of forecast accuracy and imbalance price on retailer's revenue in the electricity market.

Scenario 3 (S3): $\alpha_t^+ = 0.5$, $\alpha_t^- = 0.5$
Scenario 4 (S4): $\alpha_t^+ = 0.6$, $\alpha_t^- = 0.6$

Scenarios with different data-cost settings

Scenario 5 (S5): $\beta_{data} = 4.5 \times 10^{-5}$
Scenario 6 (S6): $\beta_{data} = 3.0 \times 10^{-5}$
Scenario 7 (S7): $\beta_{data} = 1.5 \times 10^{-5}$
Scenario 8 (S8): $\beta_{data} = 0.3 \times 10^{-5}$

The coefficients α_t^+ and α_t^- are imbalance penalty rates, which determine the economic penalty imposed on the retailer for load forecast errors. The coefficient setting covers mild-to-stringent balancing conditions. The coefficient β_{data} represents the marginal cost of offering SM data, and it is varied over more than one order of magnitude to emulate low-cost to high-cost data environments. The numerical results reported in the following section constitute a sensitivity analysis with respect to these key parameters.

5.3. The effectiveness of SM data on load forecast and the impact of forecast accuracy on electricity market revenue

This section first analyzes how SM data affects forecast accuracy and how forecast accuracy further affects the retailer's revenue in the electricity market. These analyses provide the basis for evaluating the value-cost characteristics of SM data and the subsequent two-stage benefit allocation.

To evaluate the effect of fine-grained SM data on load forecasting, two methods are compared. One method first clusters end users based on fine-grained SM data and then forecasts the load of each cluster. The other forecasts aggregate load directly using aggregated historical load data. The curve in Fig. 5 shows the forecasting error of the cluster-based aggregate load forecasting method under different fine-grained SM data volumes. The triangular reference marker on the vertical axis of Fig. 5 represents the forecasting error of the load forecasting method based on aggregated historical data. The input data of the latter method remains unchanged. Therefore, this method produces a fixed forecasting error, which is shown by the reference marker. When a sufficient volume of fine-grained SM data is available, the cluster-based aggregate load forecasting method achieves higher forecast accuracy than the load forecasting method based on aggregated historical data. Clustering is beneficial because individual household load profiles exhibit heterogeneity due to diverse appliances, occupancy patterns, and behaviors. By grouping similar users, tailored forecasting models can better capture group-specific load patterns, leading to higher aggregated forecast accuracy compared to a single model trained on heterogeneous aggregated data. The downward trend in Fig. 5 indicates that the forecast error decreases as the volume of data increases. As the main historical load patterns become increasingly represented, a further increase in data volume provides less new information for improving forecast accuracy, resulting in diminishing marginal improvement. The cluster-based aggregate load forecasting method remains effective at large data volumes, where its forecast error falls below the reference marker. In the proposed SM data market, this diminishing marginal improvement is reflected in the decreasing marginal value of SM data, as illustrated in Fig. 7. The improvement is more pronounced at small data volumes, and the marginal improvement gradually diminishes as more data is added.

After analyzing the relationship between SM data and forecast accuracy, the relationship between forecast accuracy and retailer's revenue in the electricity market is further investigated. As shown in Fig. 6, the retailer's revenue in the electricity market exhibits an approximately linear decreasing relationship with the root of MSE. A comparison across different scenarios reveals that both forecast accuracy and imbalance price significantly influence market revenue. Forecast accuracy shows a positive correlation with market revenue, whereas imbalance price shows a negative correlation. Notably, at higher imbalance price, improvements in forecast accuracy have a more pronounced effect on enhancing market revenue, indicating a higher underlying economic value.

5.4. The analysis of value-cost characteristics of smart meter data

Building on the above analysis, the value-cost characteristics of SM data are further explored. Fig. 7 illustrates the value-cost mapping relationship of SM data under varying imbalance price. In Fig. 7, the "value" refers to the increment in the retailer's expected revenue in the electricity market resulting from improved forecast accuracy. The "cost" refers to the privacy-related cost incurred by end users, as modeled in Eq. (1). Therefore, Fig. 7 provides a visual representation of the value-cost relationship that underlies the market clearing formulation. As the data volume increases, the associated privacy-related cost also increases. Meanwhile, the economic value of SM data rises because an increase in data volume improves forecast accuracy and thereby increases the retailer's revenue in the electricity market. This increase reflects the enhanced electricity market revenue derived from the high-accuracy load forecast. Notably, the economic value of SM data becomes more pronounced at higher imbalance price.

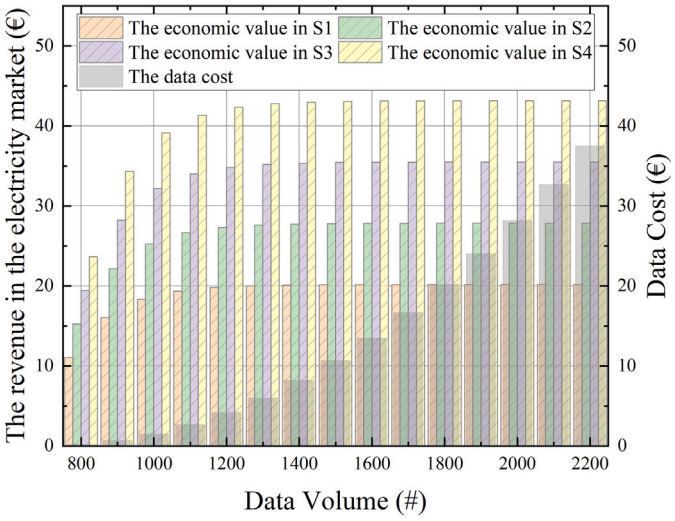


Fig. 7. The value and cost corresponding to different SM data volumes.

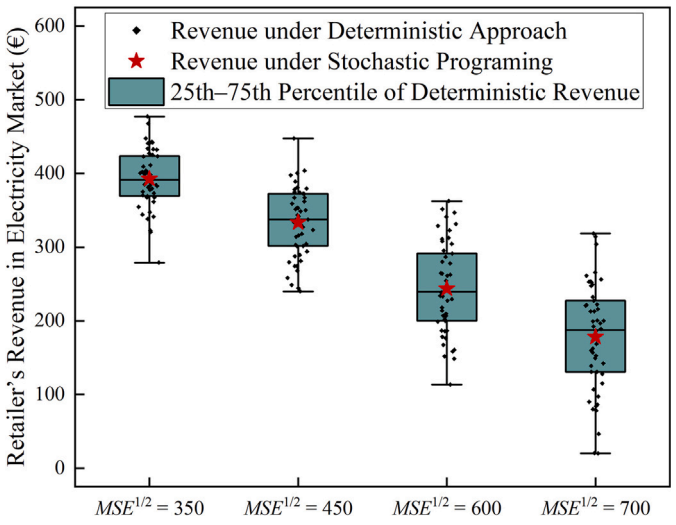


Fig. 9. Comparison of retailer's revenue under stochastic programming and deterministic single-run calculations.

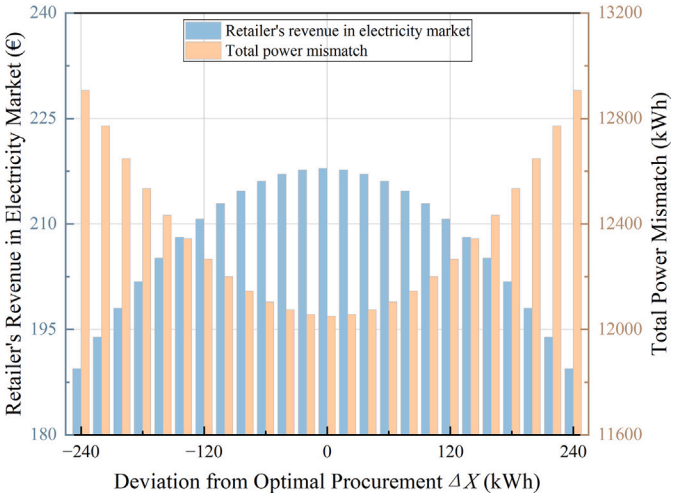


Fig. 8. Sensitivity analysis on total mismatch and retailer's revenue in the electricity market.

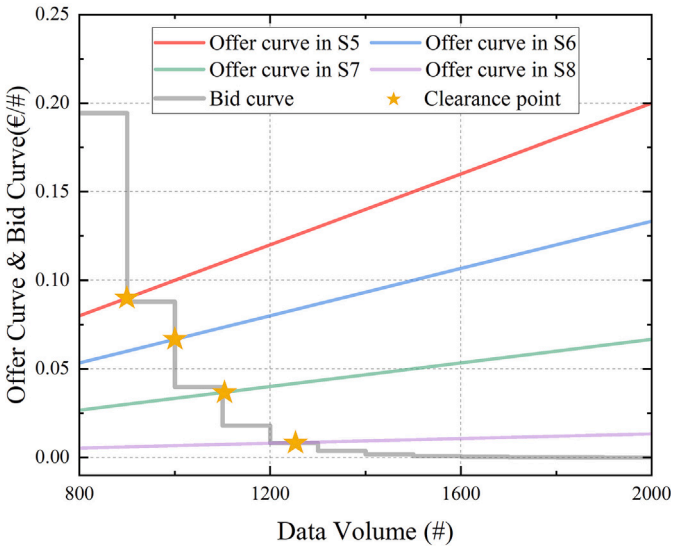


Fig. 10. SM data market clearing under different data-cost coefficient scenarios.

To validate the effectiveness of the proposed method in characterizing the economic value of SM data, a sensitivity analysis is conducted on the total power mismatch, which refers to the expected imbalance between the day-ahead electricity procurement quantity and actual load. The results are presented in Fig. 8, where ΔX denotes the deviation from retailer's optimal day-ahead procurement quantity. A positive ΔX indicates procurement above the optimal quantity, whereas a negative ΔX indicates procurement below the optimal quantity. The optimal procurement quantity is determined by Eq. (6). The blue bars represent the retailer's expected revenue in the electricity market, while the orange bars represent the total power mismatch. Each pair of bars presents the two results under one value of ΔX. The results show that deviations from the optimal day-ahead procurement quantity increase the total power mismatch and reduce the expected revenue in the electricity market. This confirms that forecast accuracy affects the retailer's revenue in electricity market through the day-ahead procurement decision, thereby supporting the value characterization of SM data.

To further clarify the role of stochastic programming, a deterministic-reference comparison is also provided. Fig. 9 compares the retailer's revenue under stochastic programming with the distribution of repeated deterministic single-run revenues under different MSE levels. The stochastic-programming result is close to the central region of the deterministic distribution, indicating that repeated deterministic calculations can serve as a reference. However, such repeated averaging is only an ex-post numerical approximation. Since the proposed framework aims to evaluate the ex-ante economic value of SM data under uncertainty, stochastic programming remains the main tool for value evaluation.

5.5. The analysis of market clearing in smart meter data market

Based on the value-cost characteristics of SM data, the market clearing results are further analyzed. The imbalance penalty rate is fixed at 0.4. The market clearing results are then analyzed under different data-cost coefficient scenarios. As shown in Fig. 10, the bid curve and offer curve represent the marginal value and marginal cost of SM data, respectively. The bid curve shows an exponentially decreasing trend, which aligns with the relationship between SM data volume and MSE shown in Fig. 5. The offer curve exhibits a linearly increasing trend because a quadratic function is used to characterize the cost of SM data in this study. Under different data-cost scenarios, the proposed market clearing mechanism achieves an outcome that maximizes social welfare. The

clearing point is reached when the marginal value of SM data equals its marginal cost. Taking the cost coefficient in Scenario 6 as an example, the results of market clearing are shown in Fig. 11.

As shown in Fig. 11, a lower data-cost coefficient leads to a larger clearing quantity of SM data and a lower clearing price. With more fine-grained SM data available, the retailer can improve forecast accuracy and increase the retailer's expected revenue in electricity market. At the same time, end-user payment decreases as the data-cost coefficient decreases. This result can be explained by the change in end users' marginal cost. A lower data-cost coefficient shifts the marginal cost curve downward, allowing more SM data to be supplied at a lower clearing price. Meanwhile, under the welfare-based market clearing formulation, the clearing price is determined by the interaction between retailer's marginal value and end users' marginal cost.

The nonlinear relationships in Eqs. (9) and (10) are reformulated by piecewise linearized approximation. After this reformulation, the market-clearing model is converted into an MIQP problem and solved by Gurobi. Such MIQP problems can be efficiently handled by commercial solvers. Since the segment number affects both approximation quality and computational burden, different segment settings are tested. The clearing result, solution time, and relative optimality gap are reported in Table 2. As shown in Table 2, the relative optimality gap reported by Gurobi remains within the prescribed tolerance for all tested segment settings. As the number of segments increases, the solution time does not increase significantly. This indicates that increasing the piecewise-linear segments does not lead to a noticeable computational burden for the optimization model. It should also be noted that the reported solution time only refers to solving the market-clearing model. Since the piecewise linearized reformulation does not affect data processing, forecasting, or parameter fitting, these procedures are not included in the reported runtime.

Table 2
Sensitivity analysis of piecewise-linear segment settings.

Segment setting ($\lvert\mathcal{N}\rvert \times \lvert\mathcal{M}\rvert$)	Q_{data} (samples)	ρ_{data} (¢/sample)	Solution time (s)	Optimality gap (%)
10×10	839.00	5.58	8.58	< 0.01
20×20	887.00	5.86	9.85	< 0.01
30×30	914.00	6.01	10.42	< 0.01
40×40	951.00	6.13	10.51	< 0.01
50×50	958.00	6.15	11.85	< 0.01
60×60	974.00	6.19	12.13	< 0.01
70×70	979.00	6.21	12.08	< 0.01
80×80	981.00	6.21	12.42	< 0.01

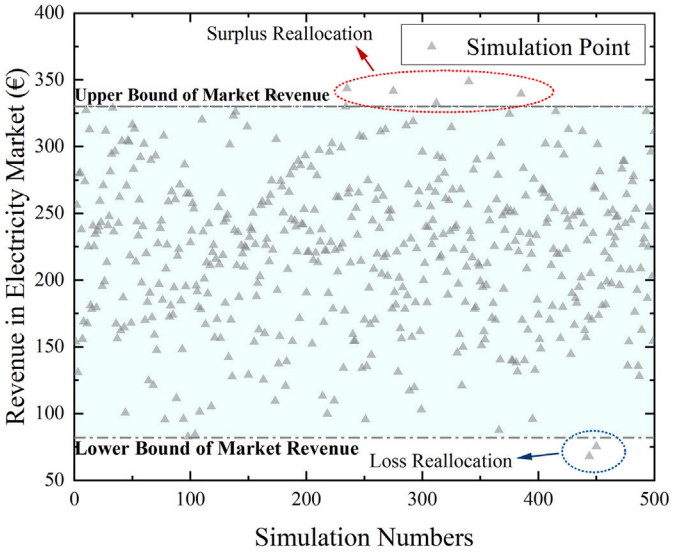


Fig. 12. Simulation of retailer's revenue in the electricity market.

5.6. The analysis of ex-post reallocation

The fluctuations in load level and electricity prices can lead to additional surplus or loss for the retailer, causing an imbalance in data market revenue. An ex-post reallocation mechanism is needed to address this issue.

Fig. 12 identifies the cases requiring ex-post reallocation by comparing the simulated retailer revenue with the allowable fluctuation range. These cases correspond to the data points outside the blue-shaded region. More points lie above the range than below it, as the proposed method adopts a conservative estimation for electricity prices when characterizing the economic value of SM data, leading to higher actual revenue. Under this method, ex-post reallocation in the surplus state occurs more frequently than ex-post reallocation in the loss state.

To verify the effectiveness of the proposed benefit reallocation method under different scenarios, several data points from Fig. 12 requiring ex-post reallocation are selected. Different allocation coefficients are set as follows: $\omega_1 = \{\alpha_{EU}^{Sur}/\beta_{RE}^{Sur} = 1,\ \alpha_{EU}^{Loss}/\beta_{RE}^{Loss} = 1\}$, $\omega_2 = \{\alpha_{EU}^{Sur}/\beta_{RE}^{Sur} = 0.5,\ \alpha_{EU}^{Loss}/\beta_{RE}^{Loss} = 0.5\}$, $\omega_3 = \{\alpha_{EU}^{Sur}/\beta_{RE}^{Sur} = 0.3,\ \alpha_{EU}^{Loss}/\beta_{RE}^{Loss} = 0.3\}$. These settings are not intended to identify a

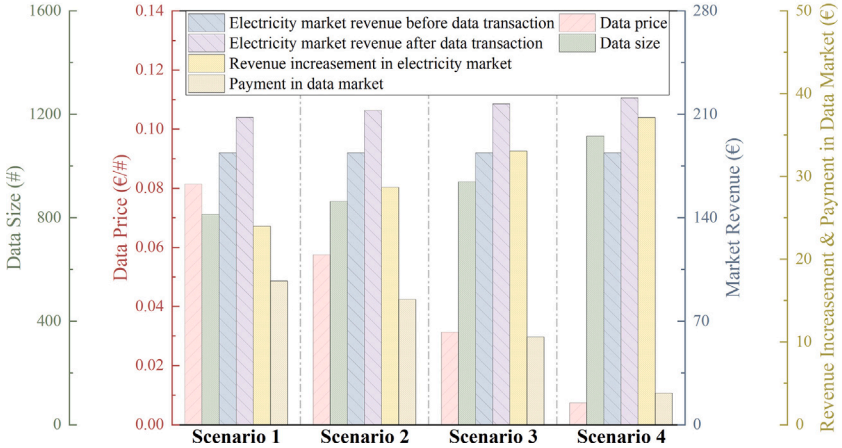


Fig. 11. SM data market clearing results and benefit allocation.

Table 3
Benefit allocation results associated with ex-post reallocation.

No.	ω	Benefit allocation of retailer (€)			Benefit allocation of end users (€)			Allocation index s^{EU}
		Initial	Ex-post	Final	Initial	Ex-post	Final	
1	ω_1	13.55	11.42	24.97	15.13	11.42	26.55	0.50
	ω_2	13.55	15.56	29.11	15.13	7.28	22.41	0.32
	ω_3	13.55	17.63	31.18	15.13	5.21	20.34	0.23
2	ω_1	13.55	8.65	22.20	15.13	8.65	23.78	0.50
	ω_2	13.55	11.86	25.41	15.13	5.43	20.56	0.31
	ω_3	13.55	13.47	27.02	15.13	3.82	18.95	0.22
3	ω_1	13.55	3.63	17.18	15.13	3.63	18.76	0.50
	ω_2	13.55	5.17	18.72	15.13	2.09	17.22	0.29
	ω_3	13.55	5.94	19.49	15.13	1.32	16.45	0.18
4	ω_1	13.55	1.47	15.02	15.13	1.47	16.60	0.50
	ω_2	13.55	2.29	15.84	15.13	0.65	15.78	0.22
	ω_3	13.55	2.70	16.25	15.13	0.24	15.37	0.08
5	ω_1	13.55	−2.82	10.73	15.13	−2.82	12.31	0.50
	ω_2	13.55	−4.09	9.46	15.13	−1.55	13.58	0.73
	ω_3	13.55	−4.73	8.82	15.13	−0.91	14.22	0.84

unique optimal coefficient combination. Instead, they are used to represent different allocation preferences between the retailer and end users. In particular, larger coefficients assigned to end users correspond to a stronger user-oriented allocation preference, whereas larger coefficients assigned to the retailer indicate a stronger retailer-oriented allocation preference.

To quantify the ex-post reallocation outcomes in a unified way for both surplus and loss cases, Table 3 reports an allocation index s^{EU}. This index is defined using the ex-post outcomes U_{EU}^{expost} and U_{RE}^{expost}, and it is normalized to $[0, 1]$.

$$s^{EU} = \begin{cases} \frac{U_{EU}^{expost}}{U_{EU}^{expost}+U_{RE}^{expost}}, & U_{EU}^{expost} \geq 0,\ U_{RE}^{expost} \geq 0, \\ 1 - \frac{|U_{EU}^{expost}|}{|U_{EU}^{expost}|+|U_{RE}^{expost}|}, & U_{EU}^{expost} \leq 0,\ U_{RE}^{expost} \leq 0, \end{cases} \tag{34}$$

Under this definition, a larger s^{EU} consistently indicates a larger relative advantage for end users. When $s^{EU} > 0.5$, end users receive a larger share of the surplus or bear a smaller share of the loss. When $s^{EU} < 0.5$, the allocation is biased toward the retailer.

Table 3 presents the benefit allocation results under different scenarios, including the initial allocation, ex-post reallocation, and final allocation outcomes. The ex-post reallocation mechanism does not alter the initial market clearing outcome. Instead, it adjusts the realized benefit distribution under different allocation coefficients, leading to different final allocations for the retailer and end users. The results show that the proposed method reallocates the surplus or loss between the retailer and end users in a controlled manner. As the coefficients vary, the allocation index s^{EU} shifts accordingly, indicating a change in how the surplus is shared or how the loss is borne. This tunability allows the mechanism to accommodate different market stages. For example, a user-acquisition stage may place greater emphasis on end-user participation, whereas a mature stage may place greater emphasis on revenue stability and long-term market operation. Therefore, the proposed method provides a flexible way to adjust the benefit allocation and support the sustainable operation of the SM data market.

6. Conclusion

This paper proposes an SM data market framework to promote the utilization of fine-grained SM data. Case studies show that fine-grained SM data improves aggregate load forecast accuracy and increases the retailer's expected revenue in electricity market. Under different market conditions, the proposed market clearing model determines the data volume and clearing price according to the value-cost characteristics of SM data. The ex-post reallocation method further reallocates realized benefits when revenue deviations occur. These results demonstrate that the proposed framework can facilitate the efficient circulation of SM data and achieve mutually beneficial economic outcomes. The proposed framework has broader implications for data-driven energy applications. More generally, it illustrates how energy-related data resources can be governed, monetized, and linked with downstream operational decision-making under market-based settings. Future work may further enrich the proposed framework in several directions. The possible extensions include strategic behavior in data trading, broader market settings with more diverse participant interactions, and multiple types of energy-related data beyond load-side SM data.

CRediT authorship contribution statement

Kang Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Investigation, Conceptualization. **Hongxun Hui:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

The steps of cluster-based load forecasting method are outlined as follows.

The main settings of the clustering-based forecasting model used in the case study are summarized in Table A.1. These settings include the clustering method, network architecture, data split, and training hyperparameters. The training samples are used for model training, while the test samples are used to evaluate forecasting performance over seven consecutive days. The selected period includes weekdays, weekends, and peak and off-peak periods. The forecasting procedure is repeated multiple times under the same setting, and the average forecast error is used for performance evaluation. As the day-ahead electricity market operation proceeds, the same evaluation process can

Algorithm 1 Aggregate Load Forecast based on Closed-Loop Clustering.

Closed-Loop Clustering

1: Collect fine-grained historical load data from N SMs.
2: Reshape SM data and select training and testing subsets.
3: Assign each user to a unique cluster, setting $K_{initial} = N$.
4: Conduct the i-th iteration (until membership in each cluster does not change or max iterations reached):
5: Aggregate the loads within each cluster k to train LSTM models ϕ_k^i.
6: Predict all cluster load profiles with each model to form error matrix.
7: Set $E_{k,j}^i = \varepsilon_k^{\phi_j^i}$, where $\varepsilon_k^{\phi_j^i}$ is the MSE on cluster k using the model trained on cluster j.
8: For each cluster, assign to the cluster minimizing forecast error for its current cluster.
9: Remove empty clusters, and refresh memberships to update the cluster information.
10: Obtain the cluster number $N_{cluster}$.

Forecast the retailer's aggregate load

11: For each of the $N_{cluster}$ clusters (indexed by n_i):
12: Aggregate the SM data by time sequence of all users in cluster n_i to obtain the historical load profile.
13: Train an LSTM load forecast model using the aggregated SM data for cluster n_i.
14: End
15: Use $N_{cluster}$ trained LSTM models for load forecast, each tailored to a specific end user cluster, and aggregate their predictions to obtain the retailer's overall load forecast.

be applied under different seasonal and operating conditions. In addition, the clustering-based forecasting model is applied to four seven-day test periods selected from spring, summer, autumn, and winter, yielding corresponding $MSE^{1/2}$ (MAPE) values of 644.86 kWh (13.83%), 492.01 kWh (11.07%), 528.53 kWh (11.63%), and 423.16 kWh (7.89%), respectively. In comparison, the baseline forecasting method based on aggregated historical data yields corresponding $MSE^{1/2}$ (MAPE) values of 847.77 kWh (15.89%), 740.90 kWh (13.05%), 701.14 kWh (13.75%), and 727.73 kWh (10.34%), respectively. These results confirm the reliability of the proposed forecasting framework across seasons, weekdays/weekends, and peak/off-peak periods.

Table A.1
Main settings of the clustering-based forecasting model.

Item	Setting
Clustering method	k-means
Number of clusters	Determined by silhouette criterion
Forecasting model	Sequence input layer → one LSTM layer (32 hidden units) → fully connected layer → regression output layer
Data split	10,000 training samples; 168 test samples
Training algorithm	Adam
Maximum epochs	100
Mini-batch size	256
Initial learning rate	0.01
Accuracy metric	MSE

Appendix B

Eq. (4) is rearranged to derive the optimal day-ahead procurement quantity. Taking period t as an example, the retailer's expected revenue can be written as follows.

$$\begin{aligned}
&\max \mathbb{E}\left[\Pi_{i,t}\left(\pi_t, \lambda_t^-, \lambda_t^+\right)\right] \\
&= \int_{X_{t,\min}}^{\tilde{X}_t} \left(p_t - \lambda_t^+\right) x_t f(x_t) \mathrm{d}x_t + \int_{X_{t,\min}}^{\tilde{X}_t} \left(\lambda_t^+ - \pi_t\right) \tilde{X}_t f(x_t) \mathrm{d}x_t \\
&\quad + \int_{\tilde{X}_t}^{X_{t,\max}} \left(p_t - \lambda_t^-\right) x_t f(x_t) \mathrm{d}x_t + \int_{\tilde{X}_t}^{X_{t,\max}} \left(\lambda_t^- - \pi_t\right) \tilde{X}_t f(x_t) \mathrm{d}x_t \\
&= \int_{X_{t,\min}}^{\tilde{X}_t} \left(p_t - \lambda_t^+\right) x_t f(x_t) \mathrm{d}x_t + \tilde{X}_t \int_{X_{t,\min}}^{\tilde{X}_t} \left(\lambda_t^+ - \pi_t\right) f(x_t) \mathrm{d}x_t \\
&\quad + \int_{\tilde{X}_t}^{X_{t,\max}} \left(p_t - \lambda_t^-\right) x_t f(x_t) \mathrm{d}x_t + \tilde{X}_t \int_{\tilde{X}_t}^{X_{t,\max}} \left(\lambda_t^- - \pi_t\right) f(x_t) \mathrm{d}x_t
\end{aligned} \tag{B.1}$$

Taking the derivative with respect to $\tilde{X}_t$.

$$\begin{aligned}
&\frac{\partial\left(\max \mathbb{E}\left[\Pi_{i,t}\left(\pi_t, \lambda_t^-, \lambda_t^+\right)\right]\right)}{\partial\left(\tilde{X}_t\right)} \\
&= \left(p_t - \lambda_t^+\right) \tilde{X}_t f(\tilde{X}_t) + \int_{X_{t,\min}}^{\tilde{X}_t} \left(\lambda_t^+ - \pi_t\right) f(x_t) \mathrm{d}x + \tilde{X}_t \left(\lambda_t^+ - \pi_t\right) f(\tilde{X}_t) \\
&\quad + \left(-p_t + \lambda_t^-\right) \tilde{X}_t f(\tilde{X}_t) + \tilde{X}_t \left(\pi_t - \lambda_t^-\right) f(\tilde{X}_t) + \int_{\tilde{X}_t}^{X_{t,\max}} \left(\lambda_t^- - \pi_t\right) f(x_t) \mathrm{d}x
\end{aligned} \tag{B.2}$$

After rearrangement, the first-order derivative can be expressed as follows.

$$\begin{aligned}
&\tilde{X}_t f(\tilde{X}_t)\left(\lambda_t^- - \lambda_t^+\right) + \left(\lambda_t^+ - \pi_t\right)\left(F(\tilde{X}_t) - F(X_{t,\min})\right) + \tilde{X}_t f(\tilde{X}_t)\left(\lambda_t^+ - \lambda_t^-\right) \\
&+ \left(\lambda_t^- - \pi_t\right)\left(F(X_{t,\max}) - F(\tilde{X}_t)\right) \\
&= F(\tilde{X}_t)\left(\lambda_t^+ - \lambda_t^-\right) + F(X_{t,\max})\lambda_t^- - F(X_{t,\min})\lambda_t^+ + \pi_t\left(F(X_{t,\min}) - F(X_{t,\max})\right)
\end{aligned} \tag{B.3}$$

Assuming that the forecast error follows a zero-mean Gaussian distribution, the interval $\left[X_{t,\min}, X_{t,\max}\right]$ is approximated by the three-standard-deviation range around the point forecast. With this approximation, the cumulative probabilities at the lower and upper bounds are expressed as follows.

$$F(X_{t,\min}) = 0, \quad F(X_{t,\max}) = 1 \tag{B.4}$$

With the corresponding boundary probabilities, the first-order derivative is expressed as follows.

$$\frac{\partial\left(\max \mathbb{E}\left[\Pi_{i,t}\left(\pi_t, \lambda_t^-, \lambda_t^+\right)\right]\right)}{\partial\left(\tilde{X}_t\right)} = F(\tilde{X}_t)\left(\lambda_t^+ - \lambda_t^-\right) + \lambda_t^- - \pi_t \tag{B.5}$$

By setting the above equation to zero, the optimal quantile condition is obtained as follows.

$$F(\tilde{X}_t) = \frac{\lambda_t^- - \pi_t}{\lambda_t^- - \lambda_t^+} \tag{B.6}$$

Therefore, the optimal day-ahead procurement quantity is obtained from the inverse cumulative distribution function at the optimal quantile.

Appendix C

The forecast errors are assumed to follow independent zero-mean Gaussian distributions. Under this assumption, the mean squared error corresponds to the variance.

$$MSE = \sigma^2 \tag{C.1}$$

As shown in Appendix B, the following relationship of boundary probabilities is adopted.

$$F_{X_t}\left(X_{t,\min}\right)=0,\quad F_{X_t}\left(X_{t,\max}\right)=1 \tag{C.2}$$

The lower and upper bounds, $X_{t,\min}$ and $X_{t,\max}$, can be approximated using the three-standard-deviation limits of the Gaussian forecast error distribution.

$$X_{t,\min}=\hat{X}_t-3\sigma=\hat{X}_t-3\sqrt{MSE} \tag{C.3}$$

$$X_{t,\max}=\hat{X}_t+3\sigma=\hat{X}_t+3\sqrt{MSE} \tag{C.4}$$

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